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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Metamind
Project / Solution Name	SkillShift MY
Name (Leader)	Sallam Maram Niyazi Abdulmawla
Email & Phone (Leader)	nazmaram948@gmail.com/ +601139774316
University Affiliation(s)	UniKL MIIT/ XMUM
Live Prototype URL	https://meta-mind-1.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/nazmaram948-creator/MetaMind/tree/dc22bc7743d20d1c0467a601fd88aa2b7edd5785

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

SkillShift MY has four layers:

(1) Frontend: a React/Vite web app with CV intake, Skills Radar, role cards, SDG explorer and slider, and a learning pathway.

(2) Backend: a FastAPI service exposing /api/* endpoints. Uploaded PDF or text CVs are read with pdfplumber and capped at 20,000 characters.

(3) Engine: Python with pandas and NumPy. CV phrases are matched to ESCO skills by a small pre-trained embedding model (sentence-transformers all-MiniLM-L6-v2, running on CPU). The engine computes skill momentum and role outlook from Anthropic Economic Index snapshots, weighted role match scores, and a 1,000-run Monte Carlo robustness estimate that varies skill momentum by up to plus or minus 20%.

(4) Data: a local SQLite database built from ESCO v1.2, O*NET 31.0, the Anthropic Economic Index, and team-curated course and SDG tables. Every skill, role, course and SDG ID returned by the API is checked against this database, and unknown IDs are blocked. No external generative-AI API is called at runtime. The app is hosted on Replit, where one application serves both the backend and the built frontend. RunPod and GPUs were not used.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Language of the backend and engine (Replit runtime)	Python Software Foundation License
FastAPI (>=0.115), Uvicorn (>=0.30), Pydantic (>=2.0), python-multipart (>=0.0.9)	REST API, request validation, CV file upload	MIT; BSD-3-Clause; MIT; Apache-2.0
pandas (>=2.2), NumPy (>=1.26), PyArrow (>=15)	Data handling, scoring and the 1,000-run Monte Carlo estimate	BSD-3-Clause; BSD-3-Clause; Apache-2.0
pdfplumber (>=0.11)	Text extraction from PDF CVs (20,000-character cap)	MIT
SQLite (Python sqlite3 module)	Local database built from the curated data	Public domain
requests (>=2.31), openpyxl (>=3.1), xlrd (>=2.0), PyYAML (>=6.0), python-dotenv (>=1.0)	Data build scripts and configuration	Apache-2.0; MIT; BSD-3-Clause; MIT; BSD-3-Clause
React (^19.3.1), Vite (^8.3.4), Recharts (3.10.1)	Frontend UI, build tool and charts	MIT
pytest (>=8.0), httpx (>=0.27), ESLint (^10.13.0)	Automated tests and linting	MIT; BSD-3-Clause; MIT
Lexend font (Google Fonts)	Interface typeface, loaded by the user's browser	SIL Open Font License 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Hugging Face Hub (Hugging Face)	Downloads the pre-trained all-MiniLM-L6-v2 model once, at build time. Not called while users use the app.	Free, public
Google Fonts (Google)	The user's browser loads the Lexend typeface. The app sends no user data.	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Replit Inc.)	Hosts the live prototype. One Replit application serves the FastAPI backend and the built React frontend. The database is rebuilt during each publish.	Replit Deployments (Autoscale), paid from the credit provided by the organisers. Region managed by Replit.
GitHub (GitHub / Microsoft)	Code repository, version control, pull requests and judge access.	Free plan / Global

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
all-MiniLM-L6-v2 (sentenc e-transformers/all-Mi niLM-L6-v2	Hugging Face (senten ce-transformers project)	Pre-trained, not fine-tuned. Matches CV phrases to ESCO skills on CPU.	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO v1.2	European Commission (https://ec.europa.eu/es co)	Public	CC BY 4.0	N
O*NET 31.0	U.S. Dept. of Labor (https://www.onetcente r.org)	Public	CC BY 4.0 / Public Domain	N
Anthropic Economic Index (Nov 2025 and May 2026 releases)	Anthropic, Hugging Face dataset Anthropic/Economic Index	Public	As stated on the dataset card	N
Course catalogue (60 courses)	Team-curated from public course pages (Coursera, edX and others)	Self-collec ted	Team-created	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
SDG mapping (42 occupations)	Team-derived from ESCO skills and UN SDG targets (sdgs.un.org)	Self-collec ted	Team-created	N
Synthetic personas and CVs	Created by the team for the demonstration	Self-collec ted	Team-created	N
DOSM / PERKESO vacancy data	Department of Statistics Malaysia, PERKESO. Prepared in the repository but not used in the current scoring.	Public	Open government data	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✔️ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic)	Writing and refactoring frontend, backend and engine code; running tests and API checks	Major
Claude (Anthropic chat assistant)	Debugging, data-check scripts, deployment guidance, documentation drafts	Moderate
Replit Agent (Replit)	Setting up the Replit run and deployment configuration (.replit)	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team chose the scoring approach and the data sources. Claude Code and Claude helped write and refactor code, and team members reviewed each change before committing. We checked the work with 54 automated tests, including a 50-run guardrail test confirming that no unverified skill, role or course ID is returned, with data-file checks, and by clicking through every screen of the live prototype. Problems we found, such as missing SDG rows, were fixed and re-tested. All forecasts are labelled Estimate, and the team takes responsibility for the submitted code and data.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✔️ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit (cloud credit of US$25 provided by the organisers)	Platform / Subscription (Replit Core)	Hosts the live prototype with Replit Deployments, and covered the cost of the deployment setup.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit hosts the live prototype, as the hackathon requires. SkillShift MY is a standard React frontend, FastAPI backend and SQLite database, so it could run on any cloud host or in a container. Replit was needed for the live deployment, not for the analytics.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

SkillShift MY has no accounts and does not ask for age, gender, NRIC, phone number or email. The demonstration uses synthetic CVs only. Pasted text or an uploaded file (PDF or text, up to 20,000 characters) is read on our Replit server, matched to skills in memory and used to compute the rankings for that request. It is not written to the database and is not sent to any third-party AI service, because no external AI API is called. Replit may keep ordinary server logs. The database holds only public taxonomies and curated tables (ESCO, O*NET, the Anthropic Economic Index, and our course and SDG tables).

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The course catalogue is small: 60 courses, 45 with verified hours, and cost and HRD Corp status are mostly unverified. Many roles have no course that covers their skill gaps, and the app says so. The SDG links are a first-pass mapping: 36 of 42 are flagged for human review, and the SDG score counts linked goals rather than measuring fit. Skill matching uses a small embedding model and can miss or mismatch skills, so the user confirms every skill. Forecasts depend on Anthropic Economic Index snapshots and our assumptions, and Malaysian demand data is not yet used in scoring. Because no generative model runs at runtime, the app cannot invent roles or courses, and every ID is checked against the database. All forecasts are labelled Estimate, and people keep the final decision.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✔️ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✔️ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✔️ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✔️ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✔️ Any use of official SDC sponsor resources has been declared in Section 5.
- ✔️ The prototype described here is accessible to judges through the links provided in Section 1.
- ✔️ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



25.10.2026